



locus (轨迹) of the midpoints of the chords is a straight line

Parabola: $y = ax^2 + bx + c$ (抛物线)

Circle: $(x-h)^2 + (y-k)^2 = r^2$

Ellipse: $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$

Hyperbola: $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$ 双曲线

$$A(x, x) + L(x) + C = 0$$

Quadratic form

Linear form

证明 locus 是 straight line = 线性

Bilinear forms: linear numerical functions of two vector arguments.

$A(x, y)$

当 $x=y$ 时 Bilinear form 变成 Quadratic form

Symmetric bilinear forms: $A(x, y) = A(y, x)$.

One of the basic problems of plane analytic geometry is to reduce the general equation of a second-degree curve to canonical form by transforming to a new coordinate system.

$$Ax^2 + 2Bxy + Cy^2 = D.$$

对称.

$$Q(x) = x^T A x$$

为什么通常只考虑 A symmetric? 因为非对称部分会抵消.

$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}$$

$$x^T (A - A^T) x = -x^T (A^T - A) x = -x^T (A - A^T)^T x = -\left(x^T (A - A^T) x\right)^T$$

$$\text{所以 } x^T (A - A^T) x = 0$$

$$Q(x) = x^T A x = \frac{1}{2} x^T \underbrace{(A + A^T)}_{\text{Symmetric}} x$$

常数 1-D-维

A is called the matrix of the quadratic form.

$\begin{cases} \text{positive definite} : Q(x) > 0 \text{ for all } x \neq 0 \\ \text{negative definite} : Q(x) < 0 \text{ for all } x \neq 0 \\ \text{indefinite} : Q(x) \text{ assumes both positive and negative values} \end{cases}$

$\begin{matrix} \text{positive} \\ Q(x) \geq 0 \end{matrix}$ semi definite -
 $Q(x) \leq 0$ negative semidefinite.

positive definite $\Leftrightarrow$ eigenvalues of A are all positive. } Theorem

Principal Axes Theorem : there is an orthogonal change of variable, $x = Py$ that transforms the quadratic form $x^T A x$ into a quadratic form $y^T D y$ with no cross-product term.

x, y = change of variable

要求 P 可逆. Change 的目的是 消除 cross product.

columns of P = basis.

y is the coordinate vector. (在新的 basis 下, 对应的新的坐标).

(y is the coordinate vector of x relative to the basis of $\mathbb{R}^n$ determined by the columns of P).

$$x^T A x = (Py)^T A Py = y^T (P^T A P) y$$

这里可以展开讲.
Coordinate System.

$\rightarrow$ New matrix of the quadratic form is $P^T A P$.
 $\rightarrow$ D : Diagonalization of Symmetric Matrices.

$\frac{1}{2} \sum y_i^2$

$\rightarrow$ diagonal matrix. 因此, 无 cross product terms.

Theorem : An $n \times n$ matrix A is orthogonally diagonalizable if and only if A is a symmetric matrix.

$$A = P D P^{-1} = P D P^T$$

研究 matrix diagonalization 的用途之一是 消除 quadratic forms 的 cross-product terms.

$$P^T A P = P^T P D P^{-1} P \quad (P^T = P^{-1})$$

$= D$ $\rightarrow$ P is an orthogonal matrix.